



The Minimum Effective Training Dose Required to Increase 1RM Strength in Resistance-Trained Men: A Systematic Review and Meta-Analysis

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Abstract

Background Increases in muscular strength may increase sports performance, reduce injury risk, are associated with a plethora of health markers, as well as exerting positive psychological effects. Due to their efficiency and effectiveness in increasing total body muscular strength, multi-joint exercises like the powerlifts, i.e.: the squat (SQ), bench-press (BP) and deadlift (DL), are widely used by active individuals as well as athletes in the pursuit of increasing strength. To date, the concept of a minimum dose, i.e. “what is the minimum one needs to do to increase 1-repetition maximum (1RM) strength?” has not been directly examined in the literature, especially in the context of the powerlifts. This review aims to explore the current available evidence around the minimum effective training dose required to increase 1RM strength in trained individuals in an attempt to enhance the practical guidelines around resistance-training as well as provide active individuals, athletes and coaches with more flexibility when designing a training protocol.

Methods One reviewer independently conducted the search in a PRISMA systematic approach using PubMed, SportDiscus and Google Scholar databases. The databases were searched with the following search terms/phrases and Boolean operators: “training volume” AND “powerlifting” OR “1RM strength” OR “powerlifters”, “low volume” AND “powerlifting” OR “powerlifting” OR “1RM strength”, “high vs low volume” AND “powerlifting” OR “1RM strength”, “minimum effective training dose 1RM”. Meta-analyses were performed to estimate the change in 1RM strength for the lowest dose group in the included studies.

Results From the initial 2629 studies, 6 studies met our inclusion criteria. All identified studies showed that a single set performed minimum 1 time and maximum 3 times per week was sufficient to induce significant 1RM strength gains. Meta-analysis of 5 studies showed an estimated increase for overall 1RM of 12.09 kg [95% CIs 8.16 kg–16.03 kg], an increase of 17.48 kg [95% CIs 8.51 kg–26.46 kg] for the SQ, and 8.25 kg [95% CIs 0.68 kg–15.83 kg] for the BP. All of the included studies contained details on most of the variables comprising “training dose”, such as: weekly and per session sets and repetitions as well as intensity of effort. Specific information regarding load (%1RM) was not provided by all studies.

Conclusions The results of the present systematic review suggest that performing a single set of 6–12 repetitions with loads ranging from approximately 70–85% 1RM 2–3 times per week with high intensity of effort (reaching volitional or momentary failure) for 8–12 weeks can produce suboptimal, yet significant increases in SQ and BP 1RM strength in resistance-trained men. However, because of the lack of research, it is less clear as to whether these improvements may also be achievable in DL 1RM strength or in trained women and highly trained strength athletes.

Registration This systematic review was registered with PROSPERO (CRD42018108911).

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Key Points

For trained men, the minimum effective training dose required to increase 1-repetition maximum (1RM) strength in the squat (SQ) and bench-press (BP) appears to be a single set of 6–12 repetitions performed with high intensity of effort at a training frequency of 2–3 times per week.

The minimum effective training dose can produce suboptimal, yet significant increases in SQ and BP 1RM strength.

It is currently less clear as to whether the minimum effective training dose can lead to similar strength improvements in deadlift (DL) 1RM strength or in trained women and highly trained strength athletes.

1 Introduction

Muscular strength is considered to be an important physical attribute for athletes as well as non-athletes. Increased muscular strength has been argued to have a number of benefits, ranging from increased athletic performance to a decreased mortality risk [1, 2]. Recent studies have argued that public health guidelines should place greater emphasis upon resistance training (RT) as it can help significantly increase muscular strength as well as improve cardiovascular fitness as a result of the typically high effort involved within resistance exercise [3].

Muscular strength can be measured in several ways, though is most commonly measured as the 1-repetition maximum (1RM) strength of an individual in a certain exercise, depending on the muscle group(s) being tested, e.g.: squat (SQ) 1RM as a means of assessing lower body strength. Different exercise modalities can be employed when looking to increase muscular strength such as 1RM strength, with resistance training being the most efficient and effective approach, especially in trained individuals [4]. Within the context of resistance training, increases in 1RM strength may or may not be optimal depending on the training approach utilised. For example, current literature shows that training to momentary failure with both lighter- and heavier loads (typically thought of as < 60% 1RM, and > 65% 1RM, respectively) can be effective in eliciting strength gains, but training with heavier loads without reaching momentary failure may lead to optimal strength increases due to the principle of specificity in this outcome [5]. This, however, highlights that, though magnitudes of improvement may differ, a variety of manipulations of resistance training variables can result in significant improvements in 1RM strength.

The concept of a “minimum–maximum effective dose”, and the potential existence of an inverted “U”-shaped curve in the dose relationship between training volume and hypertrophy, has been explored in the literature when looking at ways of optimising muscular hypertrophy adaptations [6, 7]. The concept of dose–response in resistance training has mostly been examined and debated from a “set volume” standpoint, looking at the number of sets per exercise or muscle group to induce specific physiological adaptations. It is important to note that RT dose needs to consider the number of sets as well as; repetitions per set, the load used, and the intensity of effort used—which is often considered as the proximity to momentary failure [8]. Previous reviews have attempted to further investigate the concept of “dose–response” in relation to 1RM strength by arguing for the use of absolute volume loads, relative volume loads as well as the integration of the RT-modified rating of perceived exertion (RPE) scale [9]. However, this has always happened in an attempt to find an “optimal” dose. At the moment, a consideration of the literature with respect to a critical synthesis of studies specifically looking at 1RM strength has not been conducted to determine what is the minimum effective dose required to increase 1RM strength in trained individuals. It is important to note that untrained individuals experience great increases in 1RM strength during the first months of training, thought to be mainly due to neural adaptations that occur as a result of being presented to a new training stimulus. In trained individuals, the concept of “what is the least amount of work necessary to increase 1RM strength?” is not clear as it has not yet been adequately examined. Studies and reviews comparing single- and multiple-set approaches to RT have indirectly addressed the concept but have not attempted to directly explore a possible minimum effective dose for strength in trained participants. The minimum effective training dose required to increase 1RM strength in trained individuals may provide useful knowledge to translate and improve current exercises guidelines for health, as well as for athletes and coaches across a plethora of sports, especially where strength might be a valued attribute.

A sport that defines performance solely by 1RM strength is powerlifting (PL). A powerlifter in competition has three 1RM attempts in the three powerlifts: the SQ, the bench-press (BP) and the deadlift (DL) with the goal being to achieve the highest PL total [10]. The powerlifts are also widely used as exercises to increase and assess strength and performance by many athletes as well as in research with recreational lifters (e.g. Rugby and American Football players) [11–14]. Aside from athletes, active individuals who seek to increase their overall body strength often utilise the three powerlifts as they are multi-joint exercises that are effective and efficient at increasing whole-body strength due to utilising multiple muscle groups at once [15–17]. The

increase in popularity of PL training and competition, as well as the utilisation of the powerlifts by individuals performing general strength training, has led to an increased need for scientific data around training methods that can increase 1RM strength in the powerlifts.

Considering the common usage of the three powerlifts, in particular by trained populations and athletes, previous reviews have focused on understanding what manipulation of resistance training variables might provide the optimal dose to increase 1RM strength. However, reviewing the current available evidence in an attempt to address the question of “what is the minimum a resistance-trained person can do and still increase 1RM strength?” may also be of great benefit to athletes and coaches looking to increase, or at least maintain, strength during deloading periods or periods with limited training time available. Understanding the importance of different training variables (e.g. load [%1RM] and volume) and their manipulation in ways to increase 1RM strength with the minimal dose might also help to increase flexibility of programming for athletes and coaches. Additionally, athletes competing in team sports often have to train a multitude of athletic qualities simultaneously depending on their position. For strength and conditioning coaches working with team sport athletes, understanding the minimum effective training dose required to increase muscle strength may be of great practical importance when designing training programmes. Thus, the present article aims to systematically review the available evidence regarding the minimum effective training dose required to increase 1RM strength in resistance-trained men.

2 Methods

The methodology of this systematic review was planned according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [18]. The review was registered on PROSPERO before beginning the search process in 2018 (Registration number: CRD42018108911). The study protocol is available at: https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=108911.

2.1 Inclusion Criteria

Peer-reviewed studies, published in English and available in full text, were included. There were no limitations for publishing year. The following criteria had to be met for inclusion: randomised trial, resistance training interventions lasting a minimum of 4-week manipulating dose (i.e. repetitions, sets, load, intensity of effort, etc.), a maximum of 1 working set per exercise per training session, study population consisting of healthy men with at least 1 year

of resistance training experience, at least one of the powerlifts being included in the training intervention (i.e. SQ, BP, DL) and a 1RM test used to assess strength changes on the powerlift(s) pre and post training intervention. Despite our initial inclusion criteria including study populations consisting of healthy men and women, we could not locate any studies involving trained women as participants while meeting the rest of our inclusion criteria.

2.2 Search Strategy

The literature search was performed between 10/09/2018 and 12/11/2018 by searching the following databases: PubMed, SPORTDiscus and Google Scholar. The databases were searched with the following search terms/phrases and Boolean operators: “training volume” AND “powerlifting” OR “1RM strength” OR “powerlifters”, “low volume” AND “powerlifting” OR “powerlifting” OR “1RM strength”, “high vs low volume” AND “powerlifting” OR “1RM strength”, “minimum effective training dose”. Titles, abstracts and full texts were screened individually by 1 of the 3 authors (PAK). Studies with irrelevant titles were excluded. Titles and abstracts were screened twice to ensure that no relevant studies were missed. After the title and abstract screening was completed, PAK reviewed 38 articles for a final inclusion. The reference list of any articles identified as relevant was screened for any possible articles suiting the inclusion criteria. During the review of full text articles, co-author JS was consulted to discuss any possible disagreements regarding inclusion/exclusion of any articles. PAK assessed the studies individually and thereafter compared their results.

2.3 Quality Assessment

Study quality was assessed using the Physiotherapy Evidence Database (PEDro) scale, an 11-item scale that includes both the 3-item Jadad scale and the 9-item Delphi list. The PEDro scale assesses methodological quality in randomised trials by rating them from 0 to 10 with 6 representing the threshold score for studies of high quality. The PEDro scale has been shown to be a valid tool when assessing methodological quality in randomised control trials [19].

2.4 Coding of Studies

Studies were read and individually coded by one of the investigators (PAK) for the following variables: study design, training intervention duration, descriptive information of subjects by grouping including, sex, body mass, age, resistance training experience and powerlifting experience if applicable, the powerlifts tested in each study, frequency of training each powerlift (days per week), weekly sets and

repetitions, session sets and repetitions, load, RPE rating or proximity to momentary failure using ‘repetitions in reserve’, whether participants were instructed to perform sets to momentary or volitional failure, whether there was a group or groups with higher total training volume (i.e. to be compared with the lowest dose group in the study), and whether other exercises were included in addition to any of the powerlifts.

2.5 Meta-Analysis

A meta-analysis was included retrospectively (i.e. was not part of the pre-registered systematic review protocol) to generate estimates for the change (Δ i.e. post- minus pre-intervention) in 1RM strength in the raw units (kgs) for the lowest dose groups in each of the included studies where sufficient data were reported to permit this. Meta-analysis was performed using the ‘metafor’ package in R (version 3.6.0; R Core Development Team, <https://www.r-project.org/>). Where the Δ scores for 1RMs in the lowest dose groups were reported in the original studies, these were used. Otherwise, they were calculated from the pre- and post-intervention 1RM scores reported. Similarly, the σ for the Δ scores was used as reported in the original studies or otherwise calculated assuming a within-participant pre-post correlation coefficient as none of the original studies reported this statistic. The calculation was performed as:

$$\sigma_{\Delta} = \sqrt{\sigma_{\text{pre}}^2 + \sigma_{\text{post}}^2 - (2 \times r \times \sigma_{\text{pre}} * \sigma_{\text{post}})}$$

Sensitivity analysis was used to examine the effect of range of assumed within-participant pre-post correlation coefficients ($r=0.5, 0.7$, and 0.9). The variance was taken as the square of the σ_{Δ} . A multilevel random effects meta-analysis with cluster robust variance estimation was used for estimation of Δ scores for 1RM across all of the studies and each powerlift reported in addition to separate random effects meta-analyses for each of the individual powerlifts. The meta-analyses were conducted with estimation of effect magnitude and precision in mind and so were not interpreted in the context of hypothesis testing against a point null. Heterogeneity was examined through the Q statistic and the I^2 statistic. The Q statistic assesses the statistical significance of the variability of effects within and between study groups; a significant Q statistic suggests that studies are likely not drawn from a common population. The I^2 statistic provides an estimate of the degree of heterogeneity in effects among a set of studies between 0 and 100%. I^2 values of 0–40% were not important, 30–60% moderate heterogeneity, 50–90% substantial heterogeneity, and 75–100% considerable heterogeneity [20].

3 Results

The search strategy resulted in 2629 potentially relevant articles, while 10 additional articles were identified through other sources. 2368 articles remained after any duplicates were removed. Of the 2368 articles, a total of 38 full text articles were assessed for eligibility and determined to be potentially relevant based on the information contained in the abstracts. The full text of these articles was screened and 8 studies were identified for possible inclusion in the paper. After consensus amongst the investigators, one study was excluded because it failed to meet the study design criteria. The studies by Marshall et al. [21] and Robbins et al. [22] were both based on the same participant data, something confirmed after corresponding with the primary author of the Marshall et al. [21] study, and thus only the Marshall et al. [21] study was included. Therefore, 6 studies were included for analysis (Fig. 1). Table 1 presents a summary of all the identified studies that were considered for this review. Additional study characteristics can be found in Table 2.

3.1 Powerlift(s) Included

Ostrowski et al. [23] and Schoenfeld et al. [24] included and tested the SQ and BP. Kramer et al. [25] included both the SQ and BP but only tested 1RM changes in the SQ. Similarly, Marshall et al. [20] included training and testing for the SQ only, while Rhea et al. [26] and Baker et al. [27] included training and testing for the BP only.

3.2 Strength changes

All of the identified studies reported significant increases in strength for the lowest volume group for any of the powerlifts tested. Ostrowski et al. [23] reported a SQ 1RM increase from 134 ± 28.4 to 144 ± 27.8 kg and a BP 1RM increase from 89.7 ± 11.4 to 93.3 ± 10.9 kg for the group performing a single set per week for each powerlift. Ostrowski et al. [23] did not report p values for within-group changes in SQ and BP 1RM but had defined significant effects at $p \leq 0.05$. Schoenfeld et al. [24], who also tested both the BP and SQ, reported SQ 1RM increases from 104.5 ± 14.2 to 123.4 ± 12.9 kg and BP 1RM increases from 93.6 ± 16.1 to 102.9 ± 15.2 kg for the group performing a single set per session for 3 weeks for each powerlift. Schoenfeld et al. [24] reported p values for between-group differences but did not report p values for within-group changes in SQ and BP 1RM but had defined significant effects at $p \leq 0.05$. Kramer et al. [25] reported a SQ 1RM increase from 101.9 ± 20.6 to 114.1 ± 18.7 kg for the group performing a single SQ set 2 times per week. Similarly, Marshall et al. [21] also reported a SQ 1RM increase from 149.0 ± 7.8 to 165.5 ± 9.2 kg for the

group performing a single SQ set 2 times per week. Kramer et al. [25] and Marshall et al. [21] did not report p values for within group changes in SQ 1RM but had defined significant effects at $p \leq 0.05$. Rhea et al. [26] and Baker et al. [27] only tested the BP and reported a 1RM increase from 64.2 ± 8.9 to 76.7 ± 28.0 kg and $+11.91$ kg, respectively, both for the group that performed a single BP set 3 times per week. Rhea et al. [26] and Baker et al. [27] did not report p values for within group changes in BP 1RM strength but had defined significant effects at $p \leq 0.05$.

3.2.1 Meta-Analysis of Change in 1RM Strength for Lowest Dose Groups

Of the studies included in this review, 5 were able to be included in the meta-analysis which included a total of 55 participants from the lowest dose groups in each of the included studies. Baker et al. [27] was not included due to the sum across the powerlifts of participants 1RMs being reported as opposed to individual powerlift results. Effects (k) were only available for the SQ ($k=4$) and BP ($k=3$). Sensitivity analysis of the assumptions for within-participant pre-post correlation had little impact upon the overall estimates and so results are presented for $r=0.7$. Overall, the multilevel random effects meta-analysis revealed with a

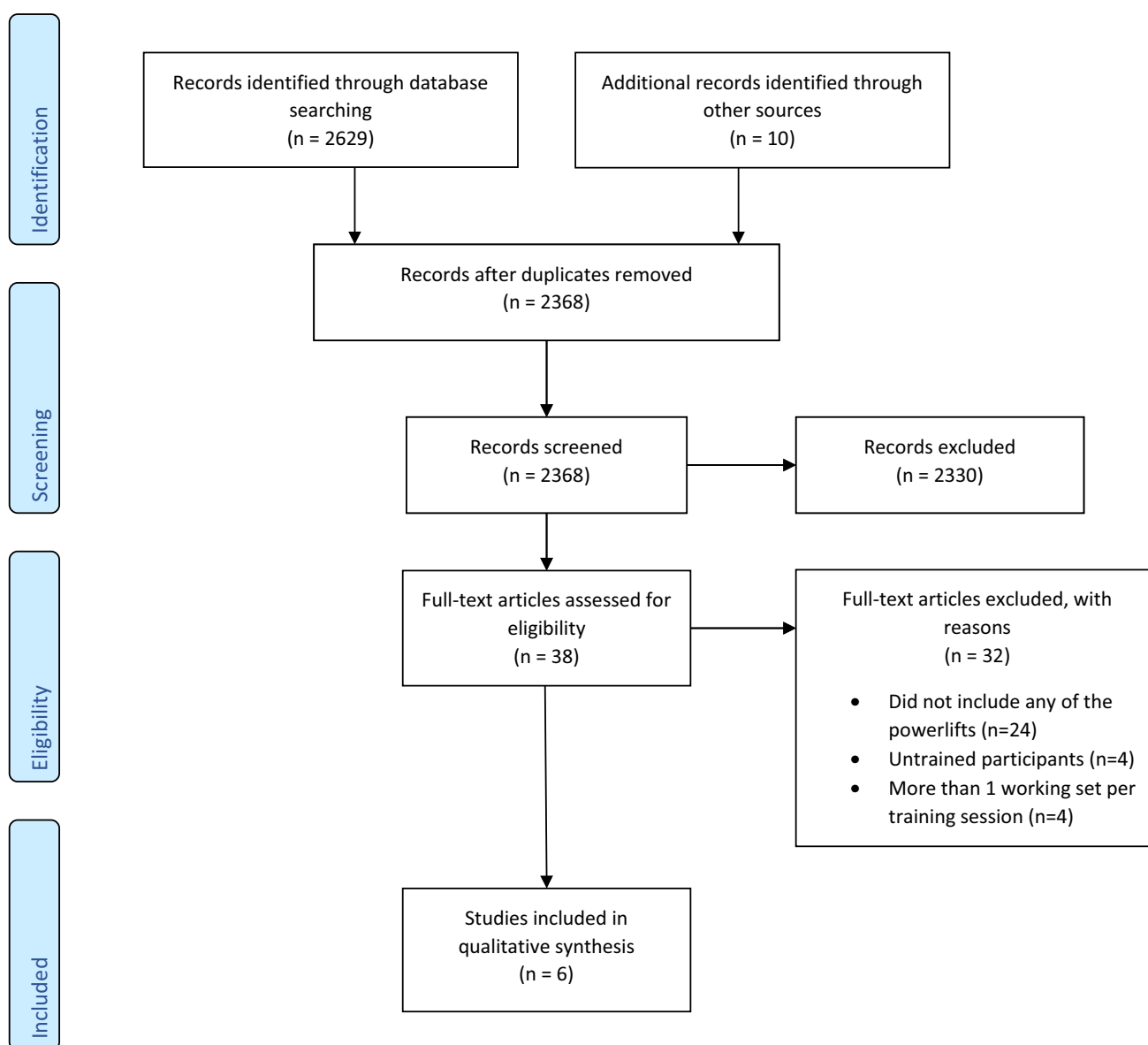


Fig. 1 PRISMA flow diagram detailing the study inclusion process

Table 1 Studies meeting inclusion criteria

Study	<i>n</i>	Design	Study duration (weeks)	Powerlift(s) tested	Sets ^a	Repetitions ^a	Weekly working repetitions ^a	Findings	PEDro score (/10)
Kramer et al. [25]	53	Random assignment to 1 of 3 training groups Single set group (SS) trained using 1 set to failure Multiple sets of 10 group (MS) trained using 3 sets of 10 reps. Multiple sets varied group (MSV) trained using multiple sets in which load and training volume varied over the training intervention	14	SQ	1	8–12	16–24	All groups experienced significant increases in SQ 1RM MS and MSV experienced approximately 50% greater 1RM strength increases than the SS group	7
Rhea et al. [26]	16	Random assignment to either a 1 set (S-1) or a 2 set (S-2) group. Both groups followed a training protocol based on daily undulating periodization (DUP) where training volume and training load varied. Subjects in both groups trained	12	BP	1	D1: 8–10 D2: 6–8 D3: 4–6	18–24	Both groups experienced significant strength increases in the BP. The S-2 group experienced significantly greater strength increases than the S-1 group	7
Ostrowski et al. [23]	35	Random assignment to 1 of 3 groups. 1 low-volume group (3 sets per muscle group per week), 1 moderate-volume (6 sets per muscle group per week) or 1 high-volume group (12 sets per muscle group per week)	10	SQ, BP	1	W1-4: ~12 W5-7: ~7 W7-10: ~9	~9	All groups experienced significant strength increases in both the SQ and BP. There were no significant differences between groups	7
Marshall et al. [21]	32	Random assignment to 6 weeks of the SQ, trained at 80% 1RM for either 1, 4 or 8 sets per week After the 6-week training period all participants followed a 4-week peaking training period	6	SQ	1	~10	~20	All groups experienced significant strength increases. The 8-set group experienced significantly more strength gains than the 1-set group. There were no significant differences between the 1-set and 4-set groups	8
Baker et al. [27]	16	Random assignment to one of two training groups. 1 group performed 1 set of upper-body exercises 3 times per week while the other group performed 3 sets of upper-body exercises 3 times per week	8	BP	1	6	18	Both training groups experienced significant strength increases. There were no significant differences between groups	7
Schoenfeld et al. [24]	34	Random assignment to 1 of 3 groups. A low-volume group (1SET) performing 1 set per exercise per session, a moderate-volume group (3SET) performing 3 sets per exercise per session or a high-volume group (5SET) performing 5 sets per exercise per session. All groups trained 3 times per week	8	SQ and BP	1	8–12	24–36	All groups experienced significant strength increases. There were no significant between-group differences	7

^aPerformed by the lower training volume group

robust estimate, an increase in 1RM of 12.04 kg [95% CIs 8.16 kg–16.03 kg]. The effect estimates from the random effects meta-analyses for individual lifts showed an increase of 17.48 kg [95% CIs 8.51 kg–26.46 kg] for the SQ, and 8.25 kg [95% CIs 0.68 kg–15.83 kg] for the BP. Q statistics were not significant for any of the analyses and heterogeneity was not evident from inspection of I^2 statistics (all 0.0%). Figure 2 shows a subgroup forest plot for the meta-analyses.

3.3 Strength Changes Compared to a Higher-Volume Group

All the identified studies included some form of comparison to a higher-volume group with some of the studies including a “moderate” volume group to assess the dose–response effect of different training volumes on strength and other measures.

Kramer et al. [25] compared 3 groups, a group performing a single set (SS) to failure, a multiple set group (MS) performing 3 sets of 10 repetitions and a multiple set varied (MSV) group performing multiple sets with varying loads and training volume. Despite all groups experiencing significant increases in SQ 1RM strength, the higher-volume groups (MS and MSV) experienced approximately 50% greater 1RM strength increases than the single-set group (SS: 101.9 ± 20.6 kg– 114.1 ± 18.7 kg compared to MS: 98.5 ± 27.7 kg– 123.7 ± 43.2 kg and MSV: 111.2 ± 25.6 kg to 135.7 ± 20.6 kg). Similarly, to Kramer et al. [25], when comparing a single-set group to group performing 3 sets, Rhea et al. [26] found that despite both groups experiencing significant strength increases in the BP, the group performing 3 sets experienced significantly greater increases when compared to the single-set group (64.21 ± 8.9 kg to 76.7 ± 28 and 66.76 ± 7.3 to 85.5 ± 20.8 , respectively).

Marshall et al. [21] compared a single-set group to a group performing 4 sets and a group performing 8 sets. Marshall et al. [21] found that all groups experienced significant strength increases and that the 8-set group experienced significantly greater strength increases than the groups performing 1 and 4 sets (1-set: 149 ± 7.8 to 162 ± 11.8 , 4-set: 157.3 ± 12.2 to 179.1 ± 11.8 , 8-set: 162 ± 11.8 to 199 ± 13.7). Interestingly, there were no significant differences between the 1-set and 4-set groups. In contrast to Marshall et al. [21], Schoenfeld et al. [24] found no significant between-group differences for 1RM strength when comparing a group performing 1 set per exercise per session to a group performing 3 sets per exercise per session or a group performing 5 sets per exercise per session (104.5 ± 14.2 to 123.4 ± 12.9 , 114.9 ± 26 to 128.5 ± 24.7 and 106.6 ± 24 to 126.2 ± 25).

Ostrowski et al. [23] compared 3 groups, a low-volume group performing 3 sets per muscle group per week, a moderate-volume group performing 5 sets per muscle group per week and a high-volume group performing 12 sets per

muscle group per week. All groups experienced significant strength increases for the SQ and there were no significant strength differences between groups (SQ: 134 ± 28.4 to 144 ± 27.8 , 146 ± 23.1 to 154 ± 20.7 and 121 ± 20.7 to 135 ± 16.3 respectively). Similarly to Ostrowski et al. [23], no significant between-group differences were observed for BP 1RM strength by Baker et al. [27] when comparing a group performing 1 set 3 times per week to a group performing 3 sets 3 times per week.

3.4 Training Intervention Length

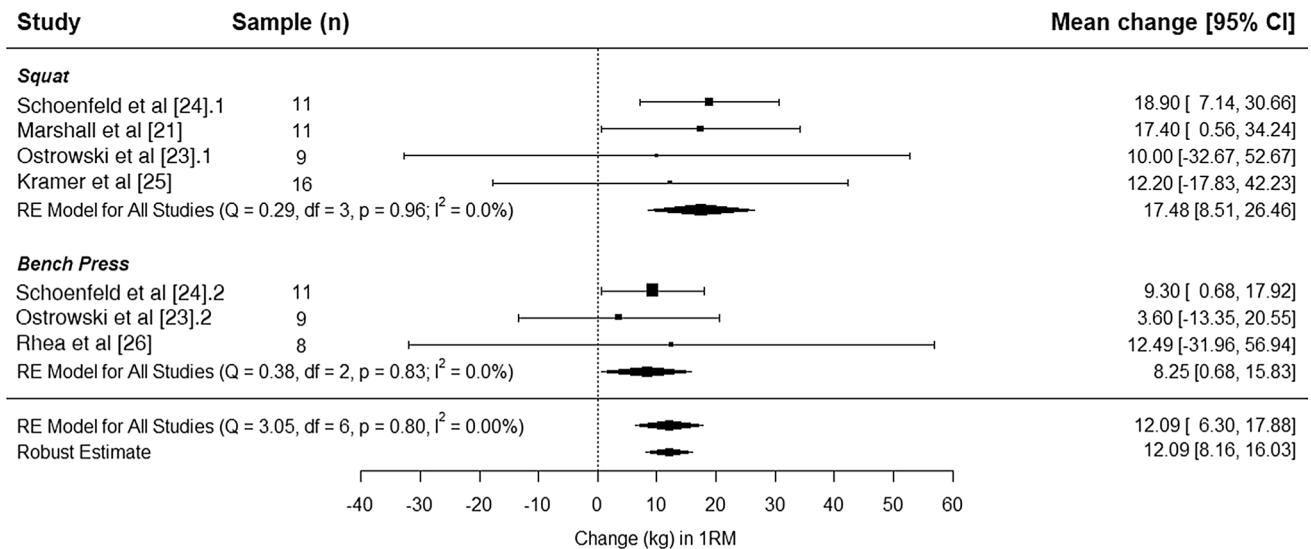
Out of the 7 identified studies, Kramer et al. [25] was the only study that exceeded 10 weeks in terms of total training intervention length. The participants in the study by Kramer et al. [25] followed a training intervention lasting 14 weeks. Participants in the studies by Marshall et al. [21] and Ostrowski et al. [23] followed a 10-week training protocol. Despite the overall length of the training intervention being 10 weeks, participants in the study by Marshall et al. [21] were divided into 1-set, 4-set and 8-set groups for 6 weeks and then followed a different training programme for the remaining 4 weeks. Marshall et al. [21] assessed 1RM strength changes at both the 6- and 10-week mark and found that participants had experienced significant strength increases at both points when compared to the baseline measurement but there were no significant differences between the 6- and 10-week mark. Participants in the studies by Baker et al. [27] and Schoenfeld et al. [24] all followed a training protocol lasting 8 weeks.

3.5 Number of Working Sets and Repetitions Per Training Session

As specified in the inclusion criteria, the low-volume group in all included studies performed a single working set [21, 23–27]. Participants in the studies by Kramer et al. [25] and Schoenfeld et al. [24] performed 8–12 repetitions per training session. Participants in the study by Ostrowski et al. [23] performed approximately 12 repetitions per training in weeks 1–4, 7 repetitions per training session in weeks 5–7 and 9 repetitions per training session in weeks 7–10. Participants in the study of Marshall et al. [21] performed an average of 10.9 ± 0.7 repetitions per training session as they were instructed to perform as many repetitions as possible until they reached momentary failure. Participants in the study by Rhea et al. [26] performed 3 training sessions per week following different repetition schemes for each session. For session 1, they performed 8–10 repetitions; for session 2, they performed 6–8 repetitions; and for session 3, they performed 4–6 repetitions.

Table 2 Additional study characteristics

Study	Age (years)	Weight (kg)	Resistance-training experience (years)	Load (%1RM)	Other exercises performed
Kramer et al. [25]	20 ± 2	80 ± 10	> 1	70–85% ^a	Yes Push-press, bench-press, crunch, pull from mid-thigh, leg curl, bent-over row
Rhea et al. [26]	21 ± 2	90 ± 12	> 2	75–90% ^a	Yes Both groups: leg press S-3: biceps curl, lat pull-down, abdominal crunches, back extensions and seated rows
Ostrowski et al. [23]	23 ± 4	77 ± 8.5	> 1	W1–4: 70% ^a W5–7: 80–85% ^a W7–10: 85–90% ^a	Yes Leg press, leg extension, stiff-leg deadlift, leg curl, single-leg curl, incline bench press, decline bench press, shoulder press, upright row, lateral raise, lat pulldown, T-bar pulldown, seated row, calf raise, calf press, seated calf raise, barbell curl, preacher curl, dumbbell curl, close grip bench, triceps push-down and triceps extension
Marshall et al. [21]	28 ± 1.2	84 ± 2.3	> 3	80%	Yes Upper-body exercises and the following during the 4-week peaking period: jump squat, bench press throw, dumbbell snatch and barbell push-press
Baker et al. [27]	20 ± 0.8	80 ± 11	> 1	85%	Yes Biceps curl and shoulder press
Schoenfeld et al. [24]	23.8 ± 3.8	82.5 ± 13.8	> 3	70–80% ^a	Yes Military press, wide grip lateral pulldown, seated cable row, machine leg press and unilateral machine leg extension

^aCorresponding load based on the NSCA loading chart [28]**Fig. 2** Forest plot of changes in 1RM (kg)

3.6 Number of Working Sets and Repetitions Per Week

Participants in the study by Kramer et al. [25] performed 2 sets and between 16 and 24 repetitions per week for the SQ.

Participants in the study by Schoenfeld et al. [24] performed 3 sets and between 24 and 36 repetitions per week for the SQ and BP. Participants in the study by Rhea et al. [26] performed 3 sets and between 18 and 24 repetitions per week for the BP. Similarly to Rhea et al. [26], participants in the study

by Baker et al. [27] performed 3 sets and approximately 18 repetitions per week for the BP. Participants in the study by Marshall et al. [21] performed 2 sets and approximately 20 repetitions per week. Finally, participants in the study by Ostrowski et al. [23] performed an average of approximately 9 repetitions per week for the SQ and BP.

3.7 Weekly Training Frequency

The weekly training frequency at which the powerlift(s) of each study were trained ranged from a minimum of 1 training session per week up to 3 training sessions per week. Participants in the studies of Kramer et al. [25] and Marshall et al. [21] trained the SQ with a training frequency of 2 times per week. One of the 3 studies that incorporated a high frequency of training for both the SQ and BP was the study by Schoenfeld et al. [24] where participants trained both powerlifts 3 times per week. Similarly, participants in the studies by Rhea et al. [26] and Baker et al. [27] trained the BP with a training frequency of 3-times per week. The lowest training frequency was seen on the study by Ostrowski et al. [23] where participants performed both the SQ and BP once per week.

3.8 Load (%1RM)

In terms of load (%1RM) used for the working sets of the studies that met the inclusion criteria, 4 out of the 6 identified studies did not report the exact load (%1RM) that was used during the training intervention [23–26]. Thus the corresponding load (%1RM) was calculated based on the 2012 National Strength and Conditioning Association loading chart [28]. Participants in the study by Ostrowski et al. [23] performed different repetitions depending the phase of the intervention they were in. The corresponding load %1RM based on the repetitions to failure that participants performed are the following: Weeks 1–4: ~70%1RM, Weeks 5–7: ~80–85%1RM and Weeks 7–10: ~85–90%1RM. Participants in the study by Schoenfeld et al. [24] performed sets of 8–12 repetitions to momentary concentric failure, implying that loads at approximately 70–80%1RM were used. Kramer et al. [25] did not specify the exact loads that were used but reported the relative intensity (%1RM) for all groups of the study showing that participants in the single-set group were trained with loads ranging from 70 to 85%1RM. Similarly to Kramer et al. [25], Rhea et al. [26] did not report the exact load(s) that were used but stated that participants used loads appropriate to their 8–10RM, 6–8RM and 4–6RM, repetition ranges that correspond to loads ranging from approximately 75–90%1RM [28]. The participants in the study by Baker et al. [27] trained with loads set at 85%1RM and similarly participants in the study by Marshall et al. [21] trained with loads set at 80%1RM.

3.9 Effort

Despite none of the studies directly measuring perceived effort (i.e. proximity to momentary failure) through the use of the modified rating of perceived effort (RPE) scale for resistance training (based on repetitions in reserve), 5 of the included studies required participants to reach momentary failure [22–25, 27] while 1 study instructed participants to reach what they termed as volitional failure [21]. Even though a specific RPE rating was not prescribed and thus there is not a direct quantifiable measure of perceived effort, the fact that participants reached either momentary or volitional failure is enough to assess the level of effort in the above training interventions. Based on the fact that either momentary or volitional failure was reached, one could argue that irrespective of not having a specific measure, most participants likely trained to a relatively high, if not maximal, perceived effort.

3.9.1 Other Exercises Performed

Participants in the studies by Marshall et al. [21], Rhea et al. [26], Baker et al. [27] and Kramer et al. [25] performed additional exercises that did not engage the musculature of the main powerlift(s) that were tested. Ostrowski et al. [23] and Schoenfeld et al. [24] both included additional exercises that engaged the musculature used in the SQ and the BP.

3.9.2 Progression

Aside from 1 study that used a specific load (80%1RM) and did not include any details on weekly or session-to-session progression [21], all other studies included some form of progression strategy depending on the training approach employed. Ostrowski et al. [23], Rhea et al. [26] and Kramer et al. [25] instructed participants to use a load appropriate to reach failure in the repetition range that was prescribed for each training session. Baker et al. [27] increased the loads used linearly as participants became stronger. Both studies did not clarify at what increments the load was increased. Finally, the load used in the study by Schoenfeld et al. [24] was increased based on the supervising researcher's assessment of what would be required to reach momentary failure after a participant had completed more than 12 repetitions prior to reaching momentary failure.

4 Quality Assessment

4.1 Overall Completeness and Applicability of Evidence

The current review included 6 randomised trials consisting of a total of 186 resistance-trained men with an average

minimum resistance training experience of 2 years. All studies included only participants that were classified as trained as well as tested and included one of the 3 powerlifts in the training intervention. The low amount of included studies does not allow for a complete examination of the concept of a “minimum effective dose for 1RM strength” but the included studies were the only available evidence that matched our inclusion criteria. To further detract from the completeness of the available evidence, there were no studies that tested and included the DL in their training intervention. The only study to date that has looked at a single set for DL 1RM strength but did not examine the DL directly and instead looked at a derivative of the DL, the Romanian DL, found significant 1RM increases after 10 weeks [29]. Out of the 6 included studies, only 2 tested and included more than 1 powerlift [23, 24]; while the remaining 4 studies tested and included either the SQ or the BP. In terms of applicability, despite all studies including participants that were classified as “trained”, their strength level would classify them as “beginners–early intermediates” based on powerlifting strength standards [10], limiting the results’ applicability to strength athletes or highly trained individuals.

4.2 Quality of the Evidence

Study quality was assessed through the use of the Physiotherapy Evidence Database (PEDro) scale. All included studies had a PEDro score higher than 6 out of 10, indicating that they were of high quality. All included studies satisfied the following criteria: eligibility criteria were specified, subjects were randomly allocated to groups, allocation was concealed and the groups were similar at baseline regarding the most important prognostic indicators. Additionally, all studies also satisfied the following criteria: measures of at least one key outcome were obtained from more than 85% of the subjects initially allocated to groups, all subjects for whom outcome measures were available received the treatment or control condition as allocated or, where this was not the case, data for at least one key outcome were analysed by “intention to treat”, the results of between-group statistical comparisons are reported for at least one key outcome, the study provides both point measures and measures of variability for at least one key outcome. The 3 criteria that were not satisfied by any of the included studies were the following: there was blinding of all subjects, there was blinding of all therapists who administered the therapy, and there was blinding for all assessors who measured at least one key outcome.

4.3 Potential Biases in the Review Process

Strengths of this review include the comprehensive search strategy of 3 databases, review of reference lists of relevant

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
Baker et al [27]	+	+	-	-	+	+	?
Kramer et al [25]	+	+	-	-	+	+	?
Marshall et al [21]	+	+	-	-	+	+	?
Ostrowski et al [23]	+	+	-	-	+	+	?
Rhea et al [26]	+	+	-	-	+	+	?
Schoenfeld et al [24]	+	+	-	-	+	+	?

Fig. 3 Risk of bias summary

reviews and reference lists of all included trials, systematic appraisal of study quality through the PEDro scale and risk of bias through the Cochrane ‘Risk of bias’ tool (Fig. 3). A noteworthy weakness of this review is that we included only randomised trials, excluding studies that offer more evidence addressing the review’s question [10], something that potentially limits the reviews’ conclusions given the very limited available data on the topic.

5 Discussion

The question that this review aimed to address is “what is the minimum a trained individual can do and still experience significant 1RM strength increases?” As mentioned above, previous studies and reviews/meta-analyses have looked at the dose–response relationship between training volume and hypertrophy as well as strength [7]. Reviews in the past have attempted to look at the dose–response relationship between training volume and strength to further understand how to

optimise training to maximize strength adaptations [30]. At present, no review has attempted to examine the current literature in an attempt to identify a possible minimum effective training dose for eliciting significant strength increases in resistance-trained individuals regardless of whether that minimum effective dose is optimal or not.

As the training dose refers to the training volume completed, it is important that training volume is operationalised at this point as its definition varies in the literature. Several studies define training volume as the absolute volume load, meaning the total tonnage completed in a specific time frame, either that being per session, week, year, etc. [25]. The main issue with absolute volume load is that all training volume is not created equally as two training protocols with equal absolute volume load may induce completely different physiological adaptations due to providing a completely different training stimulus [31]. The limitations of absolute volume load are often addressed by using relative volume load calculated by $\text{sets} \times \text{repetitions} \times \%1\text{RM}$ to account for the effect of different loads on possible adaptations. Both relative volume load and absolute volume load have the limitation of not accounting for intensity of effort per set, i.e.: 3 sets of 10 repetitions at 70%1RM versus 10 sets of 3 repetitions at 70%1RM [31], something that is not an issue in this review as all studies required participants to reach volitional or momentary failure. The issue present with the identified studies of this review is that load (%1RM) was not clearly specified or varied from week to week in 4 out of the 6 identified studies, therefore not allowing for relative volume load to be calculated. Thus, since intensity of effort was high, and therefore similar among all the studies examined, this review looked at the sets, repetitions and %1RM range as well as the total repetitions completed per session and per week in an attempt to quantify the minimum effective training dose required to increase 1RM strength in trained individuals based on the current available evidence.

Previous reviews have compared single- and multiple sets within sessions and their effects on strength and hypertrophy, with some reviews finding single sets equally effective to multiple sets [32, 33], while other reviews favouring multiple sets [34]. Despite the disagreement regarding whether multiple sets produce optimal increases in strength compared to single sets, most authors have concluded that single sets can be utilised when time is a restricting factor as they will produce strength increases. Within the present review, all studies that met the inclusion criteria utilised a single set per training session in the lower-volume group and showed significant strength increases. A single set per session seems to be enough to elicit significant strength increases in trained participants as long as certain training criteria, like intensity of effort and using relatively heavy loads (as discussed below), are met.

Prior debate regarding set volume has often placed more emphasis on discussing sets per session (either as per muscle

group or per exercise), while less emphasis has been placed on sets per week, a detail that is critical to the topic of minimum effective training dose required to increase 1RM strength. The minimum effective dose needs to be explored both from a per-session as well as a per-week standpoint, to be of practical use to individuals engaging in resistance training as well as athletes and coaches. Indeed, recent meta-analysis examining the dose–response nature of weekly volume upon 1RM strength gains has suggested that higher volumes (≥ 10 sets per week) optimise strength gains [35]. However, it should be noted that the meta-analysis of Ralston et al. still noted ‘large’ (based upon standardised effect sizes) strength gains for multi-joint exercise-specific 1RMs in the lowest weekly volume examined (≤ 5 sets per week) and only a ‘trivial’ difference between this and the highest weekly volumes. It is common among PL athletes as well as Olympic weightlifters to often look at the total amount of working repetitions that were performed at specific load zones per week per lift. A study by Gonzalez-Badillo et al. [36] investigated the effect of lower, moderate and higher volumes of training at different %1RM on experienced junior weightlifters in 10 weeks. The study found that a moderate amount of training volume per week produced superior 1RM increases in the SQ, Clean and Jerk (CJ) and Snatch (SN) exercises when compared to the lower- and higher-volume groups. Similarly, to some of the studies included in this review in adult trainees, despite being less optimal, the low-volume group still managed to experience significant 1RM increases in the SQ and CJ. The authors reported the exact training programmes followed by the participants as well as the total amount of repetitions and sets performed at each load range. When excluding the repetitions performed in the 60–70%1RM zone, which were not included in the training protocol presented in the study, the lower-volume group performed an average of 48.5 repetitions per week for the SQ. The results of their study demonstrated that it is possible to elicit significant 1RM strength increases with lower training volumes in a population that specifically trains for 1RM strength. Despite participants in the Gonzalez-Badillo et al. [36] study using a higher amount of average load when compared to the studies included in this review, they did not reach volitional or momentary failure as they performed multiple sets ranging from 1 to 6 repetitions per set. The results of the above study further support the possibility of higher effort being a potential potent variable in the context of “minimum effective training dose” since participants in the studies of this review performed less total working repetitions per week. From the studies included in this review, the highest working repetition ranges were 16–24 and 24–36 repetitions per week with loads ranging from 70 to 85%1RM [25, 27]. As briefly discussed above, recent data in PL athletes [10] showed increases in SQ and DL 1RM strength with as few as 2 and 1 repetitions (plus all warm-up

repetitions) per week performed with loads ranging from 90 to 97.5%1RM at an RPE rating of 9–9.5, hinting towards the possibility of an even lower minimum effective training dose for 1RM strength than previously thought.

As noted above, the most consistent variable that was present in all studies that were included was high effort. Participants in the included studies [21, 23–27] either reached momentary failure or what they termed volitional failure. Effort has previously been argued to be an important determinant of adaptations when investigating the effect of light loads on strength and hypertrophy [37]. Previous meta-analysis has shown that training with high amounts of effort can result in increases in both muscle hypertrophy and strength irrespective of load, as long as momentary failure is reached [38]. Similarly, the results of this review suggest that effort may be important in eliciting strength increases when training with volumes as low as 1 set per exercise per week [23, 25, 27].

The notion that low-dose training may require high efforts to produce 1RM gains is to some extent supported by a recent pilot study that was not included in the review due to not having a randomised trial study design. Androulakis-Korakakis et al. [10], compared a traditional higher-volume periodized PL protocol to an autoregulated “daily max” single-set, single-repetition protocol in powerlifters preparing for competition. Powerlifters in the “daily max” group performed a single set of a single repetition at a prescribed RPE rating of 9–9.5, meaning with a load that would not allow them to perform another repetition but was shy off the absolute maximum load that the athletes could perform on that day for a total of 10 weeks. 4 out of 5 participants experienced strength increases peri-training (i.e. during the training intervention period), while the traditional periodized PL protocol group appeared slightly better prepared for competition. The participants of the “daily max” group performed less than 10% of the total training volume than the higher-volume group but still managed to increase their peri-training PL total during the 5–7-week mark by performing 1–3 sets of a single repetition per week. Despite not reaching momentary failure, the effort of the participants was determined by the RPE rating that they were instructed to reach, resulting in a high-effort, high-load single-repetition set. Effort may be a critical variable when looking at the minimum effective training dose for 1RM strength in resistance-trained individuals. The results of the Androulakis-Korakakis et al. [10] study also indicate that perhaps if a heavier load, more specific to 1RM strength, is used then the threshold for the minimum effective training dose may be even lower than the current available data suggest.

Despite training with a high amount of effort being a possible important determinant for strength increases within the context of the review’s topic, a more important determinant for strength development is the load (%1RM) used.

Previous studies and reviews have shown that training load and not training volume may be the main determinant of 1RM strength [38] and that training loads at the 1–6RM range, which corresponds to approximately 100–85%1RM, are optimal for strength development [39]. When investigating the minimum effective training dose required to increase 1RM strength, load is one of the important variables that need to be examined. Although 4 out of the 6 included studies did not specify the exact %1RM used, the corresponding values for %1RM, based on the repetition ranges to failure reported, along with the reported load values in the remaining 2 studies, were all between the 70 and 90%1RM mark. Despite strength increases clearly being possible with light loads as long as sufficient effort is present [37], the studies included in the current review suggest that loads above the 70%1RM mark may be favourable for single sets to momentary or volitional failure for trained participants looking to increase strength by performing the minimum amount of work required. A study by Schoenfeld et al. [5] found that when comparing high-load and low-load sets to failure, both conditions resulted in significant increases in 1RM strength but heavier loads were superior as well as less time consuming. The use of heavy loads rather than light loads may also be more beneficial to strength athletes looking to train as little as possible while still addressing the element of specificity using loads that are considerably heavy or at least heavy-enough to be considered “productive repetitions” when it comes to the needs of their sport [10, 40].

Participants in 5 out of the 6 studies that met the inclusion criteria trained the powerlift(s) more than once per week resulting in a training frequency of 2–3 training sessions per lift per week [21, 24–27]. The literature on the effect of weekly training frequency, per muscle group, and its effect on 1RM strength, currently shows that a higher training frequency may potentially lead to greater 1RM gains in certain cases but not when training volume is equated [41–43]. Indeed, recent meta-analysis suggesting that low frequency training likely has similar effects to higher frequencies [44]. Despite the possibility of one single working set per week being enough to increase 1RM strength, as shown in the studies by Ostrowski et al. [23] and by Androulakis-Korakakis et al. [10] for the DL, the lack of further studies employing one single working set per week as well as the limitations of the 2 studies mentioned show that performing a single set 2–3 times a week may be required to increase 1RM strength in a particular lift. Aside from the number of weekly sets required to increase 1RM strength, managing progression from week to week can be critical to the success of any resistance training protocol and needs to be taken in consideration when looking at the minimum effective training dose. Participants in 5 out of the 6 studies reviewed, performed repetitions to muscular or volitional failure within a specified repetition range [23–27]. They progressed by

increasing repetitions until they were strong enough to surpass that prescribed repetition range, after which load was increased for the next session accounting for progressive overload. In studies where progression was not specified [21] the element of high effort allowed for progression to be “automated” by requiring participants to reach momentary or volitional failure as their strength increased throughout the training intervention. The progression method seen in the studies included in this review is of linear nature, as load or repetitions were linearly increased across each training intervention, something that is supported by the current evidence around the effectiveness of linear periodization on 1RM strength [45, 46].

Developing an understanding of how long the minimum effective training dose can be effective for is also important. All included studies were over 8 weeks long in terms of total intervention length. The recent study by Androulakis-Korakakis et al. [10], where the lower-volume group participants performed only 1–3 working repetitions per week depending on the powerlift, found that peri-training performance increased around the 5–7-week mark. Since the studies included in this review utilised sets of multiple repetitions as well as training close or to momentary failure, it is possible that the lower the weekly working repetitions are, the less time such a training approach will be effective for increasing 1RM strength. At the moment, aside from the Androulakis-Korakakis et al. [10] study, there are no data on the effect of training volume as low as a 1–3 heavy-load repetitions per week, and how long such training volume would be effective for. Interestingly, the results of this review showed that both the SQ and BP responded equally positive to similar training protocols which contrast current evidence on the lower body being more training volume dependent than the upper body [47].

An important limitation of all the studies reviewed was the inclusion of additional exercises along the powerlifts throughout the training interventions. The additional exercises in 3 of the studies included [23, 24] engaged some of the musculature involved in the main powerlift(s) tested, e.g.: leg press and shoulder press, something that could have possibly affected the strength increases experienced by the participants in the SQ and BP. It is important to note that a corresponding load calculated based on RM may not be fully accurate as different populations may be able to perform a different amount of repetitions at a given %1RM [48, 49]. Thus, the corresponding load assigned to the studies that did not report load in the form of %1RM must be interpreted with caution. An interesting limitation is that since lower repetition sets were largely not included, it is currently unknown whether training with heavier loads but fewer repetitions could be equally or even more effective than single sets of 8–12 repetitions. Another limitation was that none of the studies reviewed included the DL as part of

the powerlifts tested pre/post training intervention. Despite the minimum effective training dose possibly also being effective for DL 1RM strength, more data are required to solidify such a stance. Although participants in the studies included were resistance trained, their strength level would not be classified higher than intermediate when looked at from a PL perspective. This limits the applicability of the results of this review to highly trained strength athletes as they may not be able to increase 1RM strength with such low volumes of training. Future research should look to perform studies on strength athletes, as well as trained women, performing the 3 powerlifts and other multi-joint exercises to properly evaluate the concept of the minimum effective training dose required to increase 1RM strength. Lastly, it is unclear the extent to which the 1RM strength changes possible with low-dose resistance training, albeit significant, are practically meaningful for coaches and athletes. Our estimates suggested increases in overall 1RM of 12.04 kg [95% CIs 8.16 kg–16.03 kg], and increases of 17.48 kg [95% CIs 8.51 kg–26.46 kg] for the SQ, and 8.25 kg [95% CIs 0.68 kg–15.83 kg] for the BP. Future work should look to determine the smallest effect size that coaches and athletes would deem to be of importance (for example, using surveys) to enhance the practical applications of findings regarding strength changes from resistance training.

6 Conclusion

The results of the present systematic review suggest that performing a single set of 6–12 repetitions with loads ranging from 70–85%1RM 2–3 times per week with high intensity of effort (reaching volitional or momentary failure) for 8–12 weeks can produce suboptimal, yet significant increases in SQ and BP 1RM strength in resistance-trained men. However, because of the lack of research, it is less clear as to whether these improvements may also be achievable in DL 1RM strength or in trained women or highly trained strength athletes.

Compliance with Ethical Standards

Funding No sources of funding were used to assist in the preparation of this article.

Conflict of interest Patroklos Androulakis-Korakakis, James P. Fisher and James Steele declare that they have no conflicts of interest relevant to the content of this review.

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